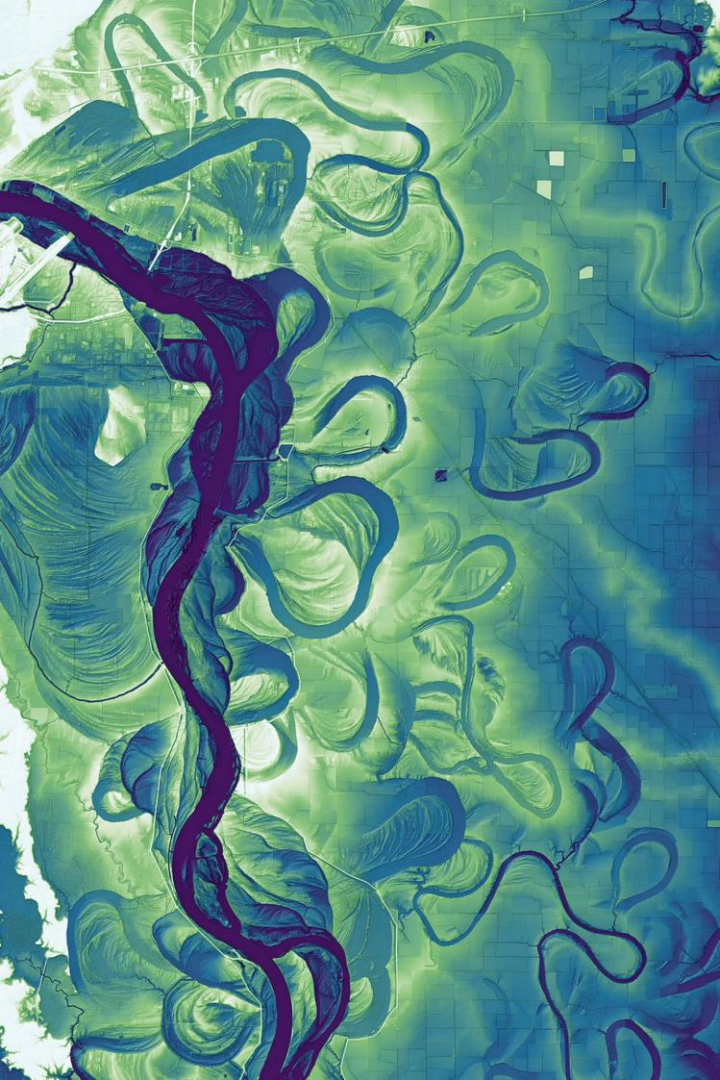




Übung II - Geologische Methoden: Vertiefung (BSc-GV05)

Introduction to GIS and Digital Cartography

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Course Roadmap

10.06.2026

Lesson 1: Map Making Basics

Intro to QGIS and creating your first map.

17.06.2026

Lesson 2: Coordinates and Projections

Understanding how we turn a round earth into a flat map.

24.06.2026

Lesson 3: Georeferencing

Converting old paper maps or images into digital data.

01.07.2026

Lesson 4: Vector Analysis

Using tools to solve spatial problems and filter data.

08.07.2026

Lesson 5: Raster Analysis

Using satellite images to calculate vegetation health (NDVI).

15.07.2026

Lesson 6: Google Earth Engine and Online Tools

Introduction to online GIS.

Why GIS? Location vs. Information

The WhatsApp Example

When you share your location on WhatsApp, you send a coordinate. It tells us **WHERE** you are.

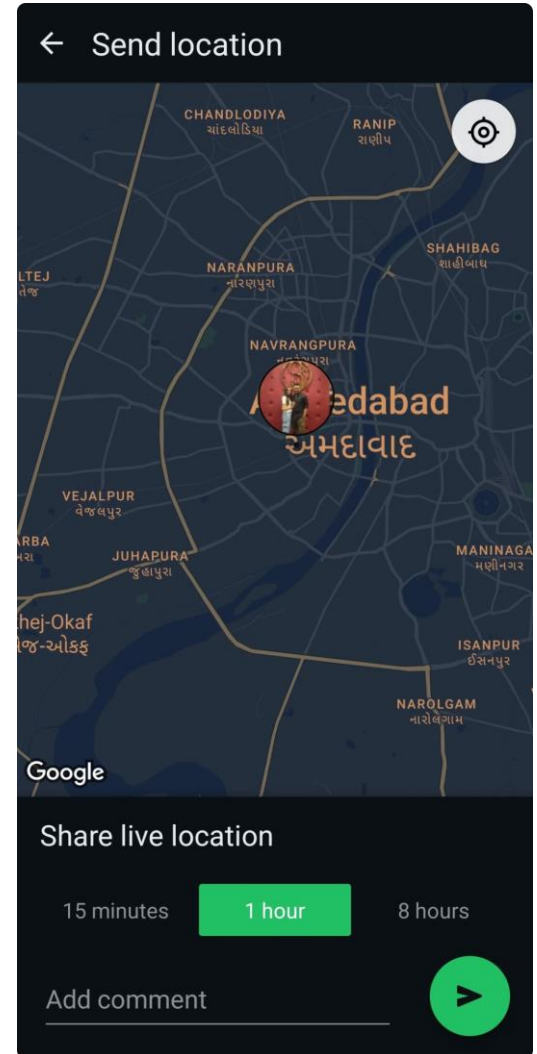
The Google Earth Example

A KML (Keyhole Markup Language) file shows a pin with a name.

The Geographic Information System (GIS) Advantage

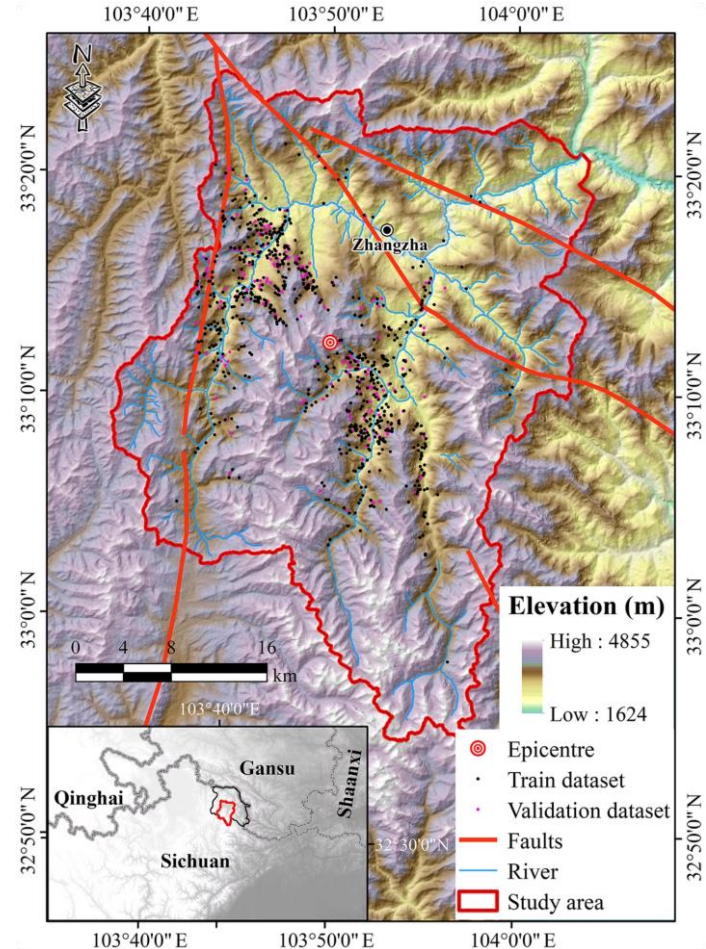
A GIS data point may contain a **whole database**. Example: For a student hometown point, we do not just see the location. We may see the student's name, their email, the population of the town, and the local geology.

 **Key Concept:** GIS is **Geometry** (the shape) plus a **Database** (the information).



Lesson 1: Map Making Basics

Objective: Create a professional map from raw data available online.



Vector vs. Raster

Data Type 1: Vector

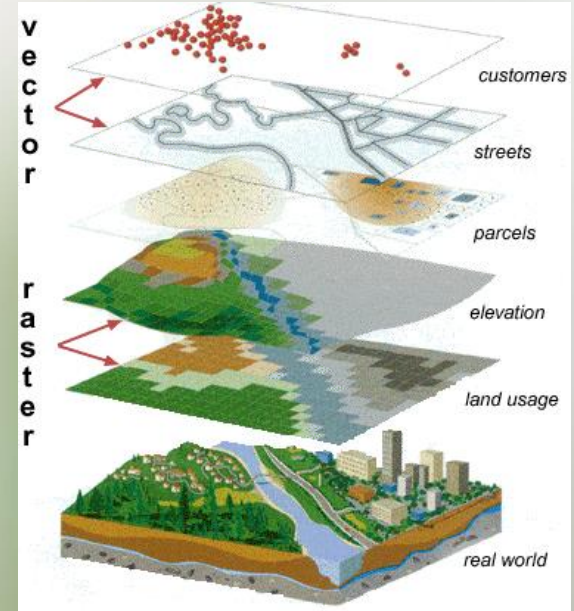
Points, Lines, and Polygons.
These are mathematical shapes.

Data Type 2: Raster

A grid of pixels. Each pixel has a value like elevation or temperature.

The *Layer Cake* Analogy

In QGIS, we usually stack these datasets. **Points** go on top, then **Lines**, then **Polygons**, and **Rasters** at the bottom.



Finding and Downloading Earth Data

- Open your web browser and go to naturalearthdata.com
- Click on **Downloads**, then select **1:10m (Large scale data)**.
- Find and download the following:

Cultural

Admin 0 - Countries (Polygons)

Cultural

Populated Places (Points)

Physical

Rivers + lake centerlines (Lines)

Raster

Cross Blended Hypso with Shaded Relief and Water (TIF)

 Unzip and copy these files into a folder on your computer named `Lesson_1/Data`. Keep your project organized.

Data Hygiene, Plugins, and Projects

The Shapefile Group

A Shapefile is a group of files (like `.shp`, `.dbf`, `.prj`). You must keep them together in one folder or the data will break.

Adding Basemaps

We can add high-quality satellite imagery using QGIS plugins. Plugins add additional functionality to QGIS.

Saving Your Work

Always save your QGIS project (`.qgz`). This file stores your styles and layer links, but it does **NOT** store the data itself.

Hints

1. Launch QGIS. In the Menu Toolbar, use **Layer → Add Layer** to find your `Lesson_1/Data` folder.
2. Drag the Raster (TIF) file into the map window first.
3. Drag the three Vector (only `.shp`) files on top of the raster.
4. Check your Layers Panel: Ensure Points are at the top, followed by Rivers, then Countries.
5. Go to the top menu: **Plugins → Manage and Install Plugins**. Search for "**QuickMapServices**" and click Install Plugin.
6. Go to **Web → QuickMapServices → ESRI → ESRI Satellite**. You now have a high-resolution satellite basemap.
7. Save your progress: Go to **Project → Save** and name your file `Lesson_1_Project.qgz` in your main folder.

Symbology and Labelling

Symbology

We will colour the countries using the `NAME` attribute. We will use **Categorised** styling so every country gets a unique colour. The rivers need to look blue as well. Let's make the countries appear *hollow*, i.e. just borders.

Labelling

We will add text to the map. We will use the `"name"` attribute to display the names of cities and rivers directly on the map.

Do the same for the countries to show their names as well.

Hints

1. Right-click the **Countries** layer → Properties → Symbology. Change the top dropdown to **"Categorised"**. Set the Value/Column to `NAME`. Click **"Classify"** at the bottom and hit OK.
2. Right-click the **Populated Places** layer → Properties → Labels. Select **"Single Labels"**. Set the Value to `"name"`.
3. Right-click the **Rivers** layer → Properties → Labels. Select **"Single Labels"**. Set the Value to `"name"`.
4. Go back to Symbology for the Rivers and change their colour to a **bright blue**. Make sure the Populated Places points are a contrasting colour so they stay visible.

The Final Product: Print Layout

Concept: The main QGIS window is for working. The Print Layout is for the final PDF/PNG.

The 5 Map Essentials



Title

What is this map showing?



Legend

What do the symbols mean?



Scale Bar

How many kilometres is one centimetre?



North Arrow

Which way is North?



Source

Data from Natural Earth.

Steps: Designing and Exporting Your Final Map

1. Go to **Project → New Print Layout**. Give it a name like "Map_1".
2. Use the **"Add Map"** icon on the left toolbar to click and drag a box on the white page. Your map will appear.
3. Use the **"Add Legend"** icon to place a legend. In the Item Properties, uncheck **"Auto Update"** to remove any layers you don't want to show.
4. Use the **"Add Scale Bar"** and **"Add North Arrow"** icons to place them in the corners.
5. Click the **"Add Label"** icon to create a text box for your Title and your Data Source (Natural Earth).
6. Once satisfied, go to **Layout → Export as PDF/PNG** to save your finished work.